U-S-746D
October 11, 1977
SUPERSEDING
Int. Fed. Spec. U-S-00746c (DSA-I
December 3, 1969 and
Fed. Spec. U-S-746b
October 24, 1957

FEDERAL SPECIFICATION

STONE, ARTIFICIAL, DENTAL

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers artificial dental stone suitable for making dental stone casts and dies.

1.2 Classification.

1.2.1 Types and sizes. Artificial dental stone covered by this specification shall be of the following types and sizes, as specified (see 6.1):

Type I - High strength stone.

Size 1. - 50 grams (24s)
Size 2. - 5 pounds
Size 3. - 35 pounds

Type II - Low strength stone.

Size 3. - 35 pounds

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

RR-S-366 - Sieve, Standard for Testing Purposes.
PPP-B-585 - Boxes, Wood, Wirebound.

FSC 6520

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PPP-B-601	-	Boxes, Wood, Cleated-plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-C-96	-	Can, Metal, 28 Gage and Lighter.
PPP-P-704	-	Pail, Metal (Shipping, Steel, 1 through 12 Gallon).
PPP-D-723	-	Drum, Fiber.

Federal Standards:

FED-STD-123	-	Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies).

Military Specifications:

MIL-P-116	-	Methods of Preservation-packaging.
MIL-L-10547	-	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards;

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing And Materials (ASTM):

C 472 - Physical Testing of Gypsum Plasters and Gypsum Concrete.

D 998 - Test for Penetration of Liquids into Submerged Containers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

Office Of The Federal Register,
National Archives and Records Service,
General Services Administration:

The Fair Practices and Labeling Act

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402)

American Dental Association (ADA):
ADA Specification No. 25 - Dental Gypsum Products.

(Application for copies should be addressed to the Council on Dental Materials and Devices, American Dental Association, 211 East Chicago Avenue, Chicago, Illinois 60611).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Dental stone shall be essentially composed of alpha hemihydrate of calcium sulfate with necessary chemical modifiers. The material shall be uniform and free of foreign material and lumps.

3.2 Description.

3.2.1 Type I. Type I, high strength artificial stone powder, when mixed with water according to manufacturer's instructions, shall be suitable for fabrication of master casts and dies.

3.2.2 Type II. Type II, low strength artificial stone powder, when mixed with water according to manufacturer's instructions, shall be suitable for fabrication, flasking and mounting of diagnostic casts.

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3.3 Physical properties. The stone shall conform to the physical properties shown in table I for the applicable type specified.

TABLE I. Physical properties.

Property	Type I	Type II
Consistency (penetration), millimeters	30 ± 2	30 ± 2
Setting time, minutes	10 ± 2	10 ± 2
Setting expansion, percent (at 2 hours)	0.00 to 0.10	0.00 to 0.20
Compressive strength (at 1 hour)	34.5 MN/m^2 (5000 psi), minimum	20.7 MN/m^2 to 34.5 MN/m^2 (3000 psi to 5000 psi)
Fineness		
Passes No. 100 sieve, minimum, percent	98	98
Passes No. 200 sieve, minimum, percent	93	93

3.4 Color. Type I stone powder shall be furnished in a white or any pale, pastel shade of color, except buff. Type II stone powder shall be furnished in a buff color only.

3.5 Weight. The stone powder shall be furnished in the weights specified in table II, when tested as specified in 4.3.9.

TABLE II. Weight of stone.

Type I	Type II
Size 1. 49-51 grams 1/ Size 2. 5-5½ pounds Size 3. 35-37 pounds	Size 3. 35-37 pounds

1/ Weight of stone in each envelope of 24 (unit of issue).

3.6 Reproduction of detail. The stone shall reproduce a groove 0.050 millimeters (mm) in width, when tested as specified in 4.3.8.

3.7 Leakage. The immediate container shall not leak when tested as specified in 4.3.10.

3.8 Lot numbers. Each container shall be marked with a serial number or combination of letters and numbers that shall refer to the manufacturer's records for that particular lot or batch. The date of manufacture (year and month) shall be given on the container either as a separate item or as a part of the lot number.

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3.9 Certification. Suppliers shall furnish at time of award a dated certificate and data showing that the dental stone they propose to furnish is the same in quality and brand as that appearing on the list of certified materials for American Dental Association Specification No. 25 for dental gypsum products. Types I and II of this specification shall be certified for types IV and III, respectively, of ADA specification No. 25. Compliance with this certification requirement does not absolve the contractor from complete compliance with the requirements of this specification.

3.10 Instructions. Adequate and accurate instructions for proportioning and manipulating the stone shall accompany each unit. Information shall include the water/powder ratio (in terms of milliliters (ml) water/100 grams (g) of powder), setting expansion and setting time. A statement that the stone shall be stored at 59° to 86° F shall be included. The instructions shall include a statement that heavy ingredients may settle to the bottom, and that for consistently accurate casts, the stone should be mixed thoroughly before dispensing. The instructions shall include a statement advising that the material should be protected from moisture contamination due to exposure to high humidity environments after the container is opened. In addition, two copies of the instructions shall be forwarded to the contracting officer for information purposes.

3.11 Delivery. No more than four months shall have elapsed from date of manufacture of the stone to date of delivery to the Government.

3.12 Workmanship. Workmanship shall be first class throughout. The stone and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.1.2 Certificates of compliance. Certificates of compliance for the requirements in 3.1 may be furnished in lieu of actual performance of such testing by the contractor. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties). Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Examination. The stone shall be examined for defects including, but not limited to, those listed in table III.

TABLE III. Classification of defects.

Categories	Defects
Major:	
101	Stone not free from foreign matter, dirt, or other impurities.
102	Stone not free from lumps.
103	Modifiers not uniformly dispersed.
Minor:	
201	Color of stone not as specified.
202	Instructions for use are inadequate, incomplete, inaccurate or missing altogether.
203	Lot number marking not as specified.

The inspection levels and acceptable quality levels (AQL's) shall be as listed in table IV. The unit of product for examination shall be one unit of issue.

TABLE IV. Examination criteria.

Type defect	Inspection level	AQL (Percent defective)
Major	S-2	1.5
Minor	S-2	2.5

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4.2.2 Testing. The stone shall be tested in accordance with table V and 4.3. If any test determination for a lot does not comply with the requirements specified, the lot shall be rejected.

TABLE V. Test criteria.

Characteristic	Requirement	Test procedure	Number of determinations per lot or batch number
Consistency	3.3	4.3.3	3 1/
Setting time	3.3	4.3.4	5 1/
Setting expansion	3.3	4.3.5	2 1/
Compressive strength	3.3	4.3.6	5 2/
Fineness	3.3	4.3.7	2 1/
Net weight	3.5	4.3.9	3 3/
Reproduction of detail	3.6	4.3.8	2
Leakage	3.7	4.3.10	2

1/ Each determination shall be the average of two specimens.
2/ Each determination shall be the average of five specimens.
3/ In process testing may be used for this characteristic.

4.2.3 Sample unit usage. Where feasible, the same unit of product shall be used for the determination of two or more test characteristics.

4.2.4 Preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in table VI. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 expressed in terms of percent defective.

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TABLE VI. Examination of preparation for delivery.

Examination	Defect
Contents	Not as specified.
Containers	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion or container cushioning inadequate, improper, or missing.

4.2.4.1 Examination of closure and reinforcement of containers. When shipping containers are required to comply with PPP-B-585, PPP-B-601, PPP-B-621, PPP-B-636, PPP-C-96 or PPP-D-723, the examination for defects in closure and banding shall be in accordance with the appendix to that specification.

4.3 Tests.

4.3.1 Test conditions. The preparation of test specimens and test procedures shall be conducted at a room temperature of 23.0 ± 2.0° C and a relative humidity of 50% ± 10%. Equipment and material shall be conditioned at this temperature and humidity at least ten hours prior to testing. Mixing equipment and molds, which must be washed before reuse on the same day, shall be brought to room temperature before each reuse.

4.3.2 Mixing Procedure. Before the stone is weighed the dry powder shall be thoroughly remixed to redistribute the ingredients properly. If there is not sufficient air space to permit free movement of the material within the immediate container, the material shall be removed from the shipping container and placed in a container (conditioned according to 4.3.1) which is capable of providing for the required free air space. The stone used in making all test specimens shall be mixed with sufficient distilled water at 23.0 ± 2.0°C to produce a mix of standard testing consistency. This consistency is defined as the water/powder ratio that will provide for a mix that will allow for a cone penetration depth meeting the requirements of 3.3. The bowl used for mixing the ingredients shall be made of rubber or plastic and shall have an open end I.D. of 10-13 cm. It shall be stored in distilled water at 23.0 ± 2.0°C for at least 10 hours before use and shall be wiped free of surface moisture immediately before adding the ingredients for mixing. The spatula used for mixing shall be a stiff bladed, round ended instrument having a blade 1.9 to 2.5 cm wide, and 9 to 13 cm long. The dry powder shall be added to the required amount of water and allowed to soak for 30 seconds. It shall then be mixed for 1 minute at the rate of 120 spatula revolutions per minute.

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4.3.3 Testing consistency by cone penetration method.

4.3.3.1 Modified Vicat apparatus. The modified Vicat apparatus (ASTM-C472) shall consist of a bracket bearing a movable metal rod 6.3 mm in diameter and of suitable length to fit the Vicat bracket. On the lower end of the rod there shall be attached a metal conical plunger having an apex angle of $53^{\circ}08'$, a height of 45 mm and a base diameter of 45 mm. The rod shall bear a mark or be equipped with a pointer to indicate movement of the rod in relation to the scale graduated in millimeters and attached to the bracket. The total weight of the rod-cone assembly shall be 35 g. The rod shall be held in a vertical position by a screw. The conical ring mold shall be made of a noncorroding, nonabsorbent material, and shall have an inside diameter of 6 cm at the base and 7 cm at the top and a height of 4 cm. The baseplate for supporting the ring mold shall be plate glass about 10 cm square. An additional weight of 65 g shall be added to the upper end of the plunger support rod when testing the consistency of dental stones. It shall be concentric with the rod and shall fit sufficiently tightly to remain positioned. A steel weight 3.5 cm long by 2 cm in diameter with a 0.63 cm hole drilled about one half the length of the weight, along the central axis, can be adjusted to give the 100 g total plunger weight.

4.3.3.2 Procedure. The following test procedure shall be used: Clean the plunger, mold and baseplate of the modified Vicat apparatus. Apply a thin coat of petroleum jelly or other suitable lubricant to the upper surface of the baseplate in order to prevent leaks during the test. The dry stone (300 g) shall be added to the test quantity of gauging liquid (1% sodium citrate in distilled water, used only for consistency determinations) and shall be mixed as described in 4.3.2. Pour this mixture into the ring mold (large end up), work slightly to remove air bubbles, and then strike off flush with the top of the mold. Wipe the conical plunger clean with a moist cloth before each determination (leave moist), and lower it to the surface of the sample. The cone shall not strike the ring mold when released. Read the scale and release the plunger quickly. After the plunger has settled, read the scale again. The difference in scale readings is millimeters of penetration. Determinations are to be made at seven, eight, and nine minutes after the start of the mix. Two penetrations are to be averaged for each determination. The average of three determinations (six penetrations) shall be taken as a measure of consistency. The water-to-powder ratio that gives the consistency specified in 3.3 shall be used for all specification testing for the stone. The release screw shall be tightened only sufficiently to hold the plunger assembly in position. Excessive force will damage the plunger support rod. The screw shall be released abruptly and completely to prevent drag on the plunger support rod and consequently low penetration readings. The mold shall be struck level with a spatula after each determination. Excess shaking, which promotes water separation, shall be avoided. Distilled water shall be used in the preparation of all specimens for the determination of setting time, setting expansion and compressive strength. Care shall be exercised to avoid contamination of testing equipment with the sodium citrate solution.

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4.3.4 Setting time. The setting time shall be determined by a standard Vicat apparatus (ASTM C472). This shall consist of a frame bearing a 300 g movable rod. One end of the rod shall be 1 cm in diameter for a distance of 5 cm; the other end shall have a removable needle 1 mm in diameter and 5 cm in length. The rod shall be reversible and it shall be held in any desired position by a screw, and shall have an adjustable indicator that moves over a scale (graduated in millimeters) attached to the frame. The sample to be tested is held in a rigid cylindrical mold that rests on a glass plate. The mold shall be made of non-corroding, nonabsorbent material and shall have an inside diameter of 2.5 cm and shall be 2.5 cm high. In addition, the Vicat apparatus shall conform to the following requirements: weight of plunger shall be 300.0 ± 0.5g; diameter of needle shall be 1.00 ± 0.05 mm. A standard mix shall be made with the use of 200 g of powder and the required amount of water. On completion, the mixed mass shall be transferred immediately to the cylindrical mold. The mold resting on the glass plate shall be completely filled, struck level, and the assembly placed on the Vicat apparatus base. The Vicat instrument plunger stop shall be adjusted such that the penetrating needle will come to rest 2.5 ± 0.5 mm from the top of the glass plate on which the specimen mold is resting. The plunger is then raised and the tip of the 1 mm needle is brought into contact with the surface of the material and locked in position. Penetrations for the trial setting time test shall begin at 2 minutes before the setting time stated by the manufacturer. The plunger shall be released to drop the needle for the first penetration at a position relative to 12 o'clock approximately 3 mm inside the test mold. Successive penetrations at 3,6,9,1,4,7 and 11 o'clock around the specimen (and at regularly spaced intervals nearer the center if necessary) shall be made at 15 second intervals until the needle fails to penetrate completely. A final penetration will then be made at the center of the specimen. The needle shall be wiped clean after each penetration. The time at which the needle fails to penetrate at the center of the specimen shall be recorded as the setting time. The total elapsed time from the start of the mix shall be reported to the nearest 0.25 minute.

4.3.5 Setting expansion. The change in length on setting shall be determined by observation of the movement of ruled markers placed approximately 30 cm apart on a specimen in a 90° V-shaped trough. These observations shall be made with a micrometer microscope comparator or by the use of equipment of equal accuracy. The V-shaped trough, into which the specimen is poured, shall be lined with rubber dam. The initial reading shall be made one minute before the time of setting as determined in 4.3.4. The final

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reading shall be made two hours after the start of mix. A standard mix composed of 300 g of the powder and the proper amount of water shall be made and shall be placed in the prepared trough. The entire mix shall be used and the overall length of the poured mass shall not exceed 36 cm. The average of two tests shall be reported to the nearest 0.01%.

4.3.6 Compressive strength. Compressive strength shall be determined on five cylindrical specimens, 20 mm in diameter and 40 mm high. Split molds shall be used to form the specimens. The prepared mix shall be poured or spatulated down the side of the inclined mold that is retained on a glass plate. The molds shall be vibrated gently while being filled. The overfilled molds shall be covered with a second glass plate that shall be rocked into place and pressed firmly into contact with the top surface of the mold. The specimens shall be removed from the split molds at one-half hour from the start of the mix and shall be stored in air at $23.0^{\circ} \pm 2.0^{\circ}C$ and 100% relative humidity until they are crushed at one hour after the start of the mix. The specimens shall be crushed at a loading rate of 300 ± 50 kilograms per minute (kg/min). Values for the five specimens shall be averaged. Values that depart from the average by more than 15% shall be discarded and the remaining values shall be averaged. If less than three values remain to be averaged, the test shall be repeated.

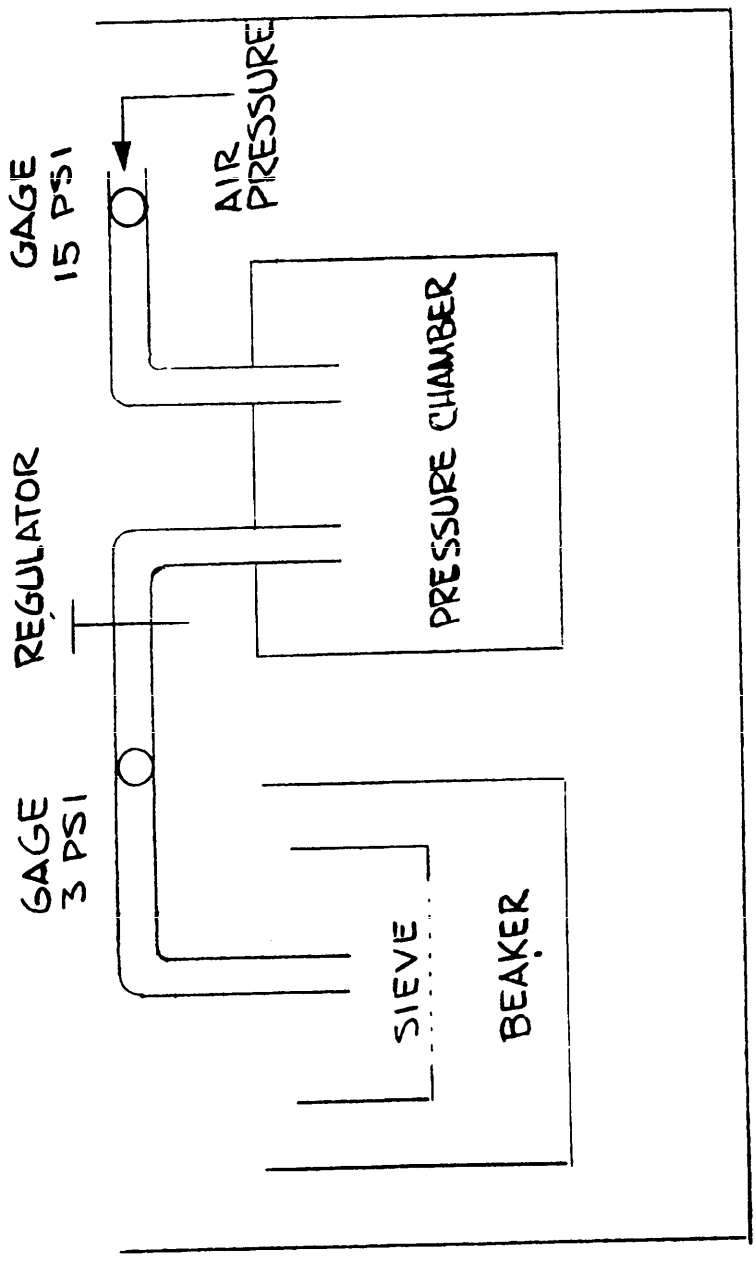
4.3.7 Fineness. Fineness shall be determined by either Method 1 or Method 2 as specified herein.

4.3.7.1 Method 1. Two samples of 100 g each shall be weighed on a balance of suitable accuracy. The first sample shall be spread uniformly over the face of a number 100 sieve, and the second over the face of a number 200 sieve. All sieves shall conform to RR-S-366. Each sample shall be rapidly and carefully washed through the respective sieve by using a swirling motion to apply a soft stream of water from a 5/8 inch inside diameter rubber hose. The water temperature shall be 18.3° to 23.9° C, and the washing operation shall be completed in 30 seconds. The sieves with remaining sample shall them be dried to constant weight in a $52^{\circ}C$ oven or screen drier. The amount of material retained on each sieve shall be weighe and the fineness calculated and expressed as a percentage of the original sample. The average values for two tests shall be reported to the nearest 0.1%.

4.3.7.2 Method 2. Fineness shall be determined by an alcohol wash method of sieving. The sieves shall conform to RR-S-366. A 10.00 g sample shall be placed on each 100 and 200 mesh sieve (Figure 1) and shall be washed with denatured alcohol at a pressure of 0.021 meganewtons per square meter (MN/m^2) (3 pounds per square inch (psi)). The sieves of 2 inches in diameter are recommended. The assembly is connected to a container, such as a pressure cooker that will hold at least 6 quarts of denatured alcohol. Air pressure is provided from any source and is reduced with a regulator to 0.103 ± 0.014 MN/m^2 (15 ± 2 psi) and then to 0.021 ± 0.002 MN/m^2 (3 ± 0.3 psi) by the assembly control. The sieve is held by hand and is rotated under the multiple stream from the nozzle. The sieve is also raised and lowered slightly while it is being rotated to insure that liquid hits the sides and all

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Procedure ... 4.3.8 Mixing



WET SIEVING APPARATUS

FIGURE 1

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parts of the screen. A 2- to 3-liter beaker is used to catch the alcohol. If the nature of the sample causes clogging of the sieve, the powder shall be partially washed through with a stream of alcohol from a wash bottle. This removes some of the fines, and wets the rest so that pressure seiving may be carried out as usual. After sieving has been carried out under pressure for 60 seconds, the sieve and the bottom of the sieve are wiped with a clean cloth or cleansing tissue. The sieve shall be transferred to a dryer at $90^0 + 5^0$C and dried for 20 minutes. When dry and cool, the sieve and the residue shall be weighed and the amount of residue shall be determined by subtracting the weight of the sieve. The average values for two tests shall be reported to the nearest 0.1%.

4.3.8 Reproduction of detail. A ring 15 mm high and with a 30 mm inside diameter shall be placed on the test block shown in figure 2 so that the intersection of a crossline and a groove 0.050 mm wide is in the center of the ring. As an alternate, a ring of the type specified above shall be positioned on a stainless steel test block similar to that shown in Figures 3 or 4 so that the three lines of the test block are centered on the ring. The prepared mix shall be poured or spatulated into the ring while the block and ring are being vibrated. The ring and the dental stone shall be separated from the test block at one-half hour and shall be examined under low-angle illumination. The reproduction of the 0.050 mm line shall be considered satisfactory if it is continuous for the full width of the ring when viewed under five power (5X) magnification.

4.3.9 Net weight. The net weight of the stone shall be determined using a balance or scale suitable for the purpose intended.

4.3.10 Leakage test. The leakage test, involving the penetration of water into the submerged immediate container, shall be determined for type I, size 1, in accordance with 5.1.1.1.1; for type I, size 2, and in accordance with 5.1.1.1.2; and for types I and II, size 3 in accordance with 5.1.1.1.3, respectively.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package.

5.1.1.1.1 Type I, Size 1. Forty-nine to fifty-one grams of stone shall be packaged in an immediate container constructed in accordance with MIL-P-116, method IC-1. Twenty-four immediate containers along with instructions for use, shall be further packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure of box shall be as specified in the appendix of PPP-B-636. Leakage test as specified in MIL-P-116, table V shall apply.

5.1.1.1.2 Type I, Size 2. Five to five-and-a-half pounds of stone along with instructions for use, shall be packaged in a hermetically sealed metal can conforming to type III, plan A of PPP-C-96. The size of the can shall not exceed 5-1/8 inches in diameter and 6-3/4 inches in height. Leakage test as specified in PPP-C-96 shall apply.

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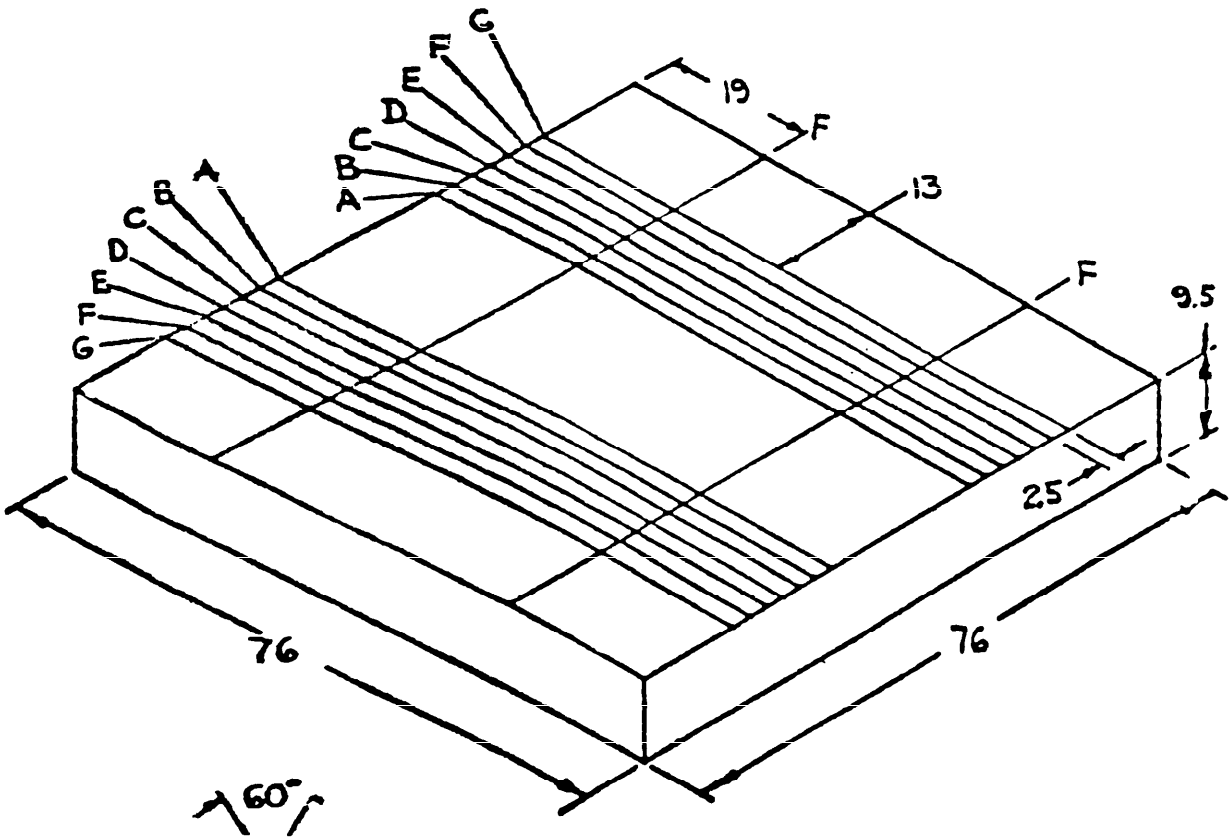
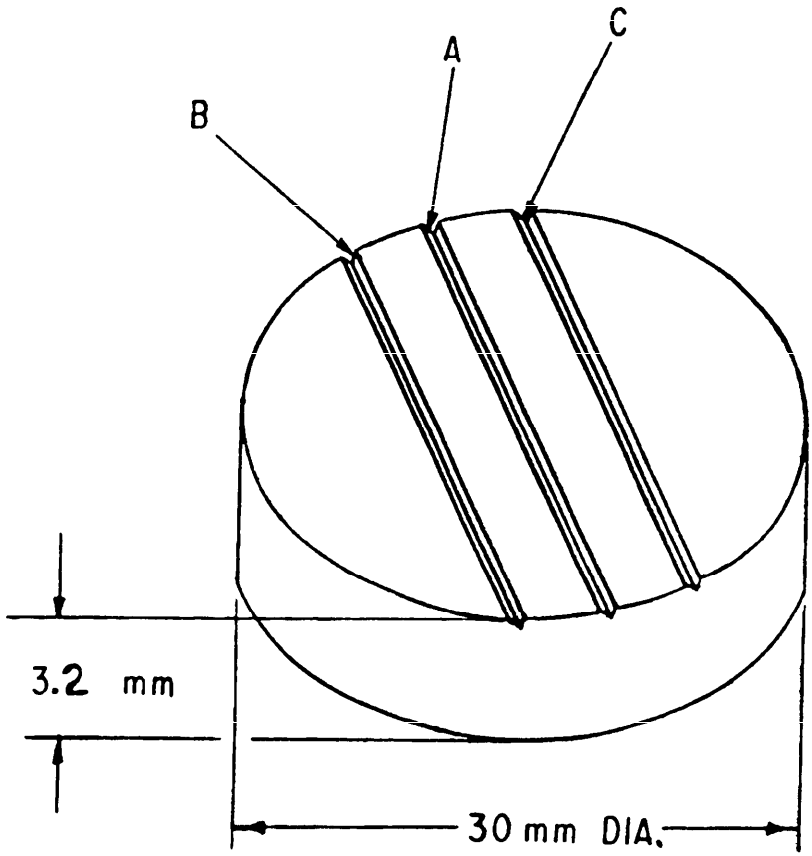
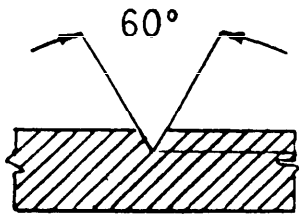


FIGURE 2. TEST BLOCK FOR DETAIL REPRODUCTION

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CROSS SECTION
OF RULED LINES

WIDTH OF RULED LINES:
LINE "A" – 0.025 mm
LINE "B" – 0.050 mm
LINE "C" – 0.075 mm

FIGURE 3
ALTERNATE TEST BLOCK FOR DETAIL REPRODUCTION

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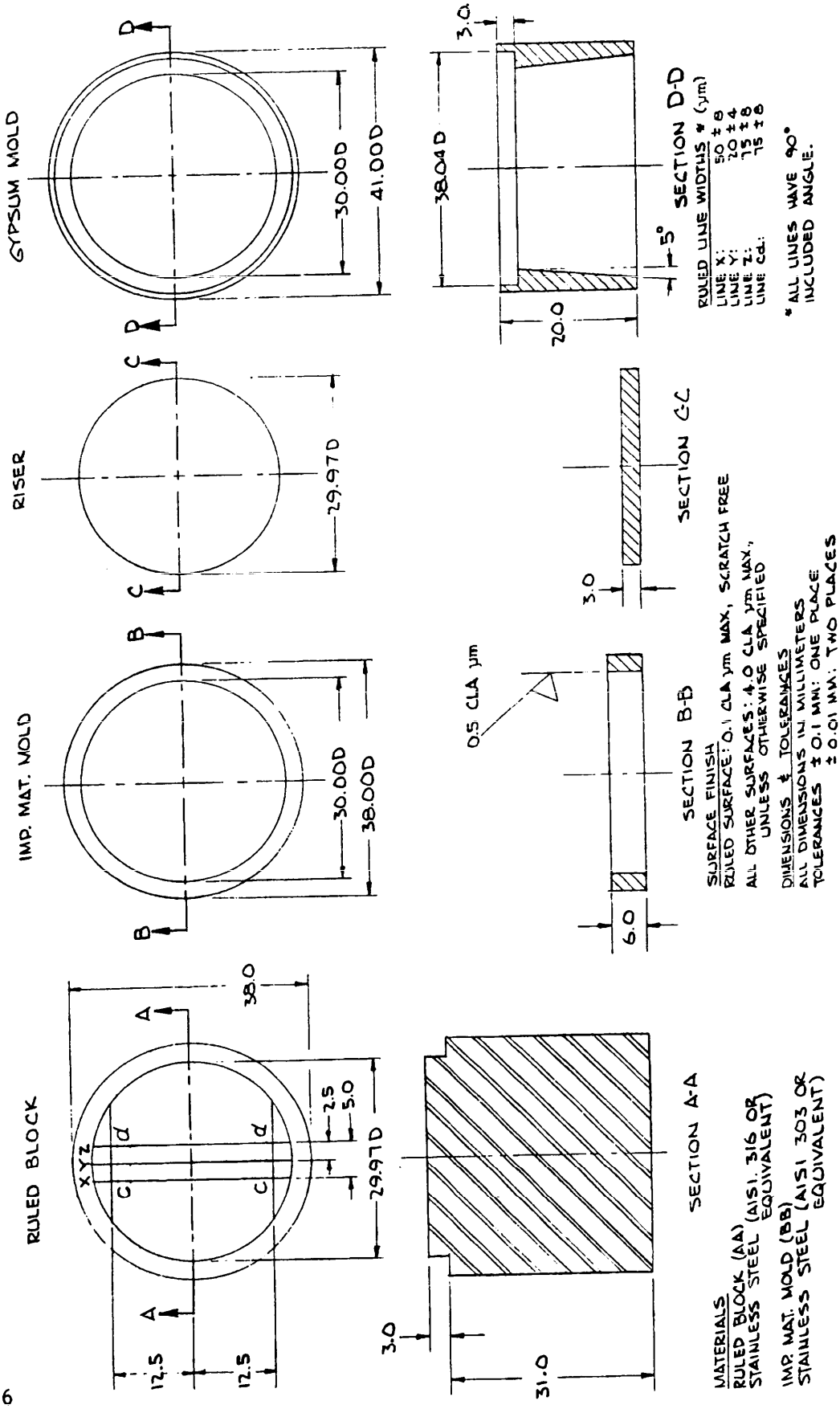


FIGURE 4 - Alternate Test Block for Detail Reproduction

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5.1.1.1.3 Type I and II, Size 3. Thirty-five to thirty-seven pounds of stone along with instructions for use, shall be packaged in a pail conforming to PPP-P-704, type II, class 1. The size of the pail shall be suitable for containing thirty-five to thirty-seven pounds of stone. Each pail shall be lined with a bag fabricated of not less than 0.002 mil polyethylene. The bag shall be secured by twisting and fastening by means of cord or metal ties. Leakage test as specified in PPP-P-704 shall apply. As an alternate, a drum conforming to PPP-D-723, Type II, grade A may be used. The drum shall be fitted with a polyethylene bag having a wall thickness of not less than 0.002 inch. Closure of the bag shall be effected by twisting and secured by means of a plastic coated metal tie. Leakage test shall not be required.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Type I, Size 1 and Type I, Size 2. Six unit packages of type I, size 1 or twelve unit packages of type I, size 2 material shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of the appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.1.2 Type I and II, Size 3. Units packaged as specified in 5.1.1.1.3 shall require no further overpacking.

5.2.2 Level B.

5.2.2.1 Type I, Size I and Type I, Size 2. Six unit packages of type I, size 1 or twelve unit packages of type I, size 2 material shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.2.2 Type I and Type II, Size 3. Units packaged as specified in 5.1.1.1.3 shall require no further overpacking.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Civil agencies. Each unit package and shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities.

5.3.2.1 Immediate container (Type I, Size 1). Each immediate container shall be commercially identified.

5.3.2.2 Unit package. Each unit package shall be marked in accordance with the Fair Practices and Labeling Act and MIL-STD-129. In addition to the Fair Practices and Labeling Act, type I and II, size 3 shall be marked as required for the exterior container specified in MIL-STD-129.

5.3.2.3 Exterior container. Each exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers artificial dental stone suitable for making dental stone casts and dies.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type and size required (see 1.2.1).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

NSN	Item Identification	Specification Designation
6520-00-116-1472	STONE, ARTIFICIAL, DENTAL, 35 lb.	Type II, Size 3
6520-00-557-7000	STONE, ARTIFICIAL, DENTAL, 5 lb.	Type I, Size 2
6520-00-557-7015	STONE, ARTIFICIAL, DENTAL, 35 lb.	Type I, Size 3
6520-00-935-1009	STONE, ARTIFICIAL, DENTAL, 50 Gram, 24s	Type I, Size 1

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Project No. 6520-1353

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DOCUMENT IDENTIFIER AND TITLE

U-S-746D STONE, ARTIFICIAL, DENTAL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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